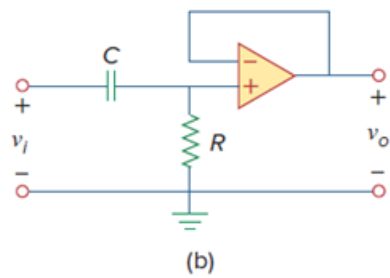
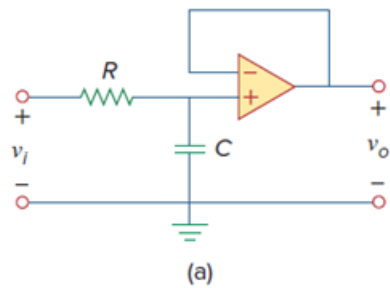


Problem

Find the transfer function for each of the active filters in Fig. 14.90.

Figure 14.90:



Step-by-step solution

Step 1 of 2

$$(a) \quad V_+ = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} V_i \quad V_- = V_o$$

since $V_+ = V_-$

$$\frac{1}{1 + j\omega RC} V_i = V_o$$

$$H(\omega) = \frac{V_o}{V_i} = \frac{1}{1 + j\omega RC}$$

Step 2 of 2

$$(b) \quad V_+ = \frac{R}{R + \frac{j}{\omega C}} V_i \quad V_- = V_o$$

since $V_+ = V_-$

$$\frac{j\omega RC}{1 + j\omega RC} V_i = V_o$$

$$H(\omega) = \frac{V_o}{V_i} = \frac{j\omega RC}{1 + j\omega RC}$$